

Local Obstructions and Slowly Growing Search Depth in the Erdős–Straus Conjecture: A Classified Census to 10^{15} and a Certified Verification to $1.8446744065 \times 10^{19}$

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Abstract

For a prime p , put $A = (p+r)/4$, $r \equiv 3 \pmod{4}$, and $q = pA$. The Erdős–Straus equation has a solution at this A exactly when a divisor of q^2 lies in the class $-q \pmod{r}$. We call the ordered search through $r = 3, 7, 11, \dots$ the *rung ladder*: for each p , we test successive rungs until the first one gives a solution. The preceding rungs have failed for that p , even though its ladder ultimately succeeds. We classify each such failure as an obstruction by the Jacobi symbol, by another real character, by the multiplicative support subgroup, or as a residual miss not explained by local support.

Among the 932,642,160,749 hard-class primes below 10^{15} , there are 557,964,077,974 unsuccessful rung tests: for each prime, we count and classify every rung before its first successful one. Jacobi certificates explain 99.1773% and residual misses account for 0.8227%. Every support-only failure occurs at $r = 51$. The identity $4A = p + r$ explains this concentration: it forces 4 into every support subgroup, a condition invisible to real characters. The maximum first-hit rung is 155. A proposed fit $M(X) = C \log X$ has the correct scale but does not reproduce the long record plateaus.

At a fixed squarefree rung, Selberg–Delange theory gives an explicit $c_r/\sqrt{\log X}$ law for the unrestricted factor-support model, while $\chi_r((p+r)/4) = \chi_r(p)$ shows why the actual ladder is not independent. These results separate a tractable local model from the cross-rung dependence still missing from a proof.

An independent certified sieve verifies every prime below $1.8446744065 \times 10^{19}$. Across 460 shards, 156,377,459,709 primes reached the individual search, all were solved, and no survivor remained. This is about 18.4 times the previous record of 10^{18} , but not the full 64-bit range: the final 8,589,936,640 integers below 2^{64} lie outside the computation.

1 Introduction

The Erdős–Straus conjecture [?] asserts that for every integer $n \geq 2$ the equation $4/n = 1/x + 1/y + 1/z$ has a solution in positive integers; it suffices to treat prime $n = p$. Polynomial congruence identities resolve the equation outside a thin set of residue classes. Mordell’s

*Computations performed with machine assistance; all numerical claims are reproducible from the released code and data (§??). Code, data, and discussion are hosted at <https://github.com/Attitude-Adjuster/erdos-s Strauss>; questions, corrections, and reproducibility reports may be filed through the repository’s issue tracker.

identities leave six classes $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$, all of them squares modulo 840 [?]. We denote their union by the *hard class* $\mathcal{H}$. The commonly used surrogate $p \equiv 1 \pmod{24}$ is four times larger and mixes these primes with classes already settled by an identity, diluting the measured residual share. For that reason, all counts below are restricted to $\mathcal{H}$. The obstruction is structural: Schinzel and Yamamoto showed that no finite list of polynomial congruence identities covers a quadratic-residue class [?, ?].

Fixing A and setting $r = 4A - p$, $q = pA$, solvability is equivalent to the existence of a divisor of q^2 in the class $-q \pmod{r}$ (Lemma ??, classical). A rung can fail for two quite different reasons:

- *Local residue support.* The divisors of q^2 all lie in some proper subset of $(\mathbb{Z}/r)^\times$ determined by the residues of the primes of q , and the target class lies outside it. A short certificate proves that the rung cannot work.
- *A residual miss.* The target class passes every local test, but none of the finitely many divisors of q^2 reaches it.

The first case can be decided entirely by arithmetic in a finite group. The second is the part left unexplained by local information. We formalize this distinction as a nested hierarchy of certificates, use it to classify every failed rung for $p < 10^{15}$ in $\mathcal{H}$, and release certificates that can be checked independently. The parametrization, character expansion, and Type I/II taxonomy are classical. Our contribution is the hierarchy, a structural theorem restricting its finest local level to an explicit finite list of rungs (§??), the exhaustive classification, a fixed-rung hazard calculation which retains the exact dependence between p and $(p+r)/4$, and the resulting picture of the slowly growing residual tail.

The paper follows a single arc. Section ?? assembles what is already known — Mordell’s identities and the covering family — into one statement: every classical formula prescribes a rung and a witness divisor in advance. Section ?? dissolves that finite list into a geometric family, the hyperbola of solutions at a fixed rung, and Section ?? turns the hyperbola’s divisor lattice into a finite Fourier expansion whose vanishing modes are exactly the certificate hierarchy. Section ?? measures the hierarchy exhaustively below 10^{15} . Section ?? then runs the arc in reverse: the same classical formulas, algorithmically implemented in a certified covering sieve, verify the conjecture to $1.8446744065 \times 10^{19}$. The final sections describe the computation, place it against prior work, and state the analytic obstacle that remains.

2 What is known, as one family

The classical results on the conjecture are usually presented as a list of separate identities. Read in the right coordinates they are one construction, and stating it once sets up everything that follows.

Mordell’s identities [?] are polynomial solutions of $4/p$ valid on residue classes of p ; together they settle every class modulo 840 except the six square classes defining $\mathcal{H}$ in the introduction. Behind them sits the covering family of Rosati [?], catalogued at scale by Salez [?]: for positive integers a, c, d, k with $ck > a$, put $m = 4acd - 1$ and $e = ck - a$; then every integer $p = mk - 4a^2d$ satisfies $p + k = 4ade$ and

$$\frac{4}{p} = \frac{1}{ade} + \frac{1}{acdp} + \frac{1}{cdep}. \quad (1)$$

Nothing about primality is used: one triple (a, c, d) solves the *entire arithmetic progression* $p \equiv -4a^2d \pmod{m}$ above an explicit floor. The family is self-certifying — the two-line verification of (??) is the certificate for every member at once.

The unifying reading is this. Writing $A = ade$, the identity gives $4A = p + k$, so each formula fixes *in advance* both the shifted value A (equivalently, the rung k of the ladder introduced in §??) and a divisor witnessing the solution. A classical identity is a single precomputed point on one of the hyperbolas of §??. Two consequences shape this paper. Constructively, the formulas can be implemented algorithmically: enumerated, certified, and swept over ranges, which is how §?? verifies the conjecture to $1.8446744065 \times 10^{19}$. Structurally, they cannot finish the job: no finite list of polynomial congruence identities covers a quadratic-residue class $[\cdot, \cdot]$, and §?? exhibits the obstruction concretely — the family (??) can never produce a perfect square. On the six square classes modulo 840, solvability must therefore be established prime by prime, and the question becomes not *whether* a formula applies but *how far down the ladder* each prime must climb. This paper focuses on that ladder.

3 The rung ladder and the divisor lattice

Fix a prime $p \equiv 1 \pmod{4}$ and an integer $A > p/4$, and set $r = 4A - p > 0$, $q = pA$. Subtracting $1/A$ from $4/p$ gives $1/B + 1/C = r/q$, and completing the product turns this into a hyperbola:

$$\frac{4}{p} = \frac{1}{A} + \frac{1}{B} + \frac{1}{C} \quad \Longleftrightarrow \quad (rB - q)(rC - q) = q^2. \quad (2)$$

Lemma 3.1 (divisor criterion; classical). *Positive integer solutions (B, C) of (??) correspond bijectively to divisors $u \mid q^2$ with $u \equiv -q$ and $q^2/u \equiv -q \pmod{r}$, via $B = (u + q)/r$, $C = (q^2/u + q)/r$. If $\gcd(q, r) = 1$ the second congruence follows from the first.*

Proof. If (B, C) solves (??) then $u = rB - q$ divides q^2 and is $\equiv -q \pmod{r}$, as is its cofactor; conversely such a u yields integers $B, C \geq 1$. When $\gcd(q, r) = 1$, $u \equiv -q$ gives $q^2/u \equiv q^2(-q)^{-1} \equiv -q \pmod{r}$. $\square$

As A runs over integers exceeding $p/4$, r runs over $r \equiv -p \equiv 3 \pmod{4}$. We call the resulting search order a *ladder* with *rungs* $r = 3, 7, 11, \dots$ and $A = (p + r)/4$; rung r *hits* if Lemma ?? produces a solution, and the *depth* of p is the smallest hitting r .

For every rung considered in the computation, q and r are coprime.

Lemma 3.2. *If $0 < r < 3p$ then $\gcd(q, r) = 1$.*

Proof. $r < 3p$ implies $A < p$, so $p \nmid A$ and, p being prime, $\gcd(A, p) = 1$. Then $\gcd(A, r) = \gcd(A, 4A - p) = \gcd(A, p) = 1$ and $\gcd(p, r) = \gcd(p, 4A - p) = \gcd(p, 4A) = 1$ since p is odd and $p \nmid A$. Hence $\gcd(pA, r) = 1$. $\square$

Lemma ?? shows that the second congruence in Lemma ?? adds no further condition in our computation. Since r/q is reduced, the criterion also has a classical two-unit-fraction form and a centered group-theoretic form.

Proposition 3.3 (Rav's criterion in three forms). *Let $\gcd(q, r) = 1$ and write $q = \prod_{\ell} \ell^{e_{\ell}}$. The following are equivalent.*

1. *Some $u \mid q^2$ satisfies $u \equiv -q \pmod{r}$.*

2. Some coprime pair of divisors $d_1, d_2 \mid q$ satisfies $d_1 + d_2 \equiv 0 \pmod{r}$.
3. The fixed class -1 belongs to the centered exponent box

$$B_{q,r} := \left\{ \prod_{\ell} \ell^{f_{\ell}} \pmod{r} : f_{\ell} \in \mathbb{Z}, |f_{\ell}| \leq e_{\ell} \right\} \subseteq (\mathbb{Z}/r)^{\times}. \quad (3)$$

Proof. We pass between the three formulations as follows. Given the pair in (2), put $t = (d_1 + d_2)/r$, $B = qt/d_1$ and $C = qt/d_2$. These are integers and $1/B + 1/C = r/q$. Moreover $u = rB - q = qd_2/d_1$ is a divisor of q^2 , its complementary divisor is qd_1/d_2 , and $u \equiv -q \pmod{r}$.

Conversely, given u in (1), set $g = \gcd(u, q)$, $d_1 = q/g$ and $d_2 = u/g$. If $\ell^e \parallel q$ and $\ell^j \parallel u$, with $0 \leq j \leq 2e$, then

$$v_{\ell}(d_1) = \max(0, e - j), \quad v_{\ell}(d_2) = \max(0, j - e).$$

Thus d_1 and d_2 are coprime divisors of q . Because g is invertible modulo r , we can cancel it from $gd_2 = u \equiv -q = -gd_1 \pmod{r}$ and obtain $r \mid d_1 + d_2$. The two constructions are inverse; swapping d_1, d_2 corresponds to replacing u by q^2/u .

Finally, $u/q = d_2/d_1$ has exponent vector $f_{\ell} = v_{\ell}(d_2) - v_{\ell}(d_1) \in [-e_{\ell}, e_{\ell}]$. As u ranges over the divisors of q^2 , these are exactly the vectors in (??), and $u \equiv -q \pmod{r}$ becomes $u/q \equiv -1 \pmod{r}$. $\square$

Remark 3.4 (attribution and role). Form (2) is Rav's classical criterion for a reduced fraction m/n to be a sum of two unit fractions, here with $m/n = r/q$ [?]; sometimes this criterion is attributed to Schinzel, and was communicated to us in this form, by Elsholtz [?]. Chen and Koo state the criterion in this form [?]. Form (1) is the completed hyperbola (??). Form (3) is the group-theoretic reparametrization used below: the support subgroup H of Definition ?? is generated by the same residues after the exponent bounds are removed. Levels J, RC and S say that the subgroup already misses -1 ; level R says that the subgroup contains -1 but the finite box does not.

The divisors of $q^2 = p^2 A^2$ form a box lattice: writing $A = p^e A'$ with $p \nmid A'$ and $A' = \prod_i \ell_i^{a_i}$, they are $u = p^j d$ with $0 \leq j \leq 2e + 2$ and $d \mid A'^2$. Reducing modulo r colors this grid, and Lemma ?? asks whether the color $-q$ appears. The classical two-type taxonomy is the row coordinate:

Proposition 3.5 (types are rows). *Suppose $p \nmid A$ and $\gcd(q, r) = 1$, and let a solution arise from $u = p^j d$ with $j \in \{0, 1, 2\}$ and $d \mid A^2$. If $j = 1$ then $p \mid B$ and $p \mid C$ (Type II); if $j \in \{0, 2\}$ then p divides exactly one of B, C (Type I).*

Proof. $p \mid q$, so $p \mid u + q$ iff $p \mid u$, i.e. iff $j \geq 1$; the cofactor has p -exponent $2 - j$. Since $\gcd(r, p) = 1$, $p \mid B$ iff $p \mid u + q$. $\square$

Remark 3.6. For $p = 73$, $A = 20$ ($r = 7$, $q = 1460$) the lattice contains four solution divisors $u \leq q$: $u = 10, 80$ in row $j = 0$ (Type I, e.g. $4/73 = 1/20 + 1/210 + 1/30660$) and $u = 73, 584$ in row $j = 1$ (Type II, e.g. $4/73 = 1/20 + 1/219 + 1/4380$). Both classical families sit in one lattice.

4 The character expansion and a hierarchy of certificates

Throughout, $\gcd(q, r) = 1$ and $N_r(p) = \#\{u \mid q^2 : u \equiv -q \pmod{r}\}$, counting all divisors (each unordered solution $\{B, C\}$ contributes two).

Theorem 4.1 (divisor count in a class; orthogonality). *With $q^2 = \prod_\ell \ell^{a_\ell}$,*

$$N_r(p) = \frac{1}{\varphi(r)} \sum_{\chi \pmod{r}} \overline{\chi(-q)} \prod_{\ell \mid q} (1 + \chi(\ell) + \chi(\ell)^2 + \cdots + \chi(\ell)^{a_\ell}). \quad (4)$$

Proof. For u, a coprime to r , orthogonality [?, Ch. 4] gives $\varphi(r)^{-1} \sum_\chi \chi(u) \overline{\chi(a)} = \mathbf{1}[u \equiv a]$. Take $a = -q$, sum over $u \mid q^2$, and factor the completely multiplicative χ across the divisor structure of q^2 . $\square$

The conjugate in (??) is essential: omitting it counts the class $(-q)^{-1}$ rather than $-q$. Both evaluations were checked against direct enumeration; the examples are included in the repository tests.

The trivial character contributes $\tau(q^2)/\varphi(r) > 0$; the others contribute cancellation. For real characters that cancellation is itself a divisor count:

Lemma 4.2. *Let χ be a real character mod r . Since every exponent a_ℓ in q^2 is even,*

$$\sum_{u \mid q^2} \chi(u) = \prod_{\chi(\ell)=1} (a_\ell + 1) = \tau((q^2)_+) \geq 0,$$

where $(q^2)_+$ is the part of q^2 supported on primes with $\chi(\ell) = 1$: at primes with $\chi(\ell) = -1$ the local factor $1 - 1 + \cdots + 1$ equals 1.

The obstruction applies to every odd r and every real character.

Corollary 4.3 (real-character certificate). *Let r be odd with $\gcd(q, r) = 1$ and let χ be any real Dirichlet character mod r . If $\chi(\ell) = +1$ for every prime $\ell \mid q$ and $\chi(-q) = -1$, then $N_r(p) = 0$.*

Proof. χ is completely multiplicative, so every divisor $u \mid q^2$ has $\chi(u) = +1$, whereas χ is constant on residue classes mod r and equals -1 at $-q$. Hence no divisor lies in that class. Equivalently in (??): by Lemma ?? the χ term is $\chi(-q)\tau(q^2) = -\tau(q^2)$, cancelling the trivial term exactly. $\square$

Remark 4.4 (the Jacobi symbol is only one of them). For odd $r = \prod_{i=1}^\omega p_i^{e_i}$ the group $(\mathbb{Z}/r)^\times$ has exactly 2^ω real characters, namely the products $\chi_T(x) = \prod_{i \in T} \left(\frac{x}{p_i}\right)^{e_i}$ over subsets $T \subseteq \{1, \dots, \omega\}$. The Jacobi symbol is one of them—but *not* in general the full-subset one: $\left(\frac{\cdot}{r}\right) = \chi_{T_0}$ with $T_0 = \{i : e_i \text{ odd}\}$, since $\left(\frac{x}{p_i^{e_i}}\right) = \left(\frac{x}{p_i}\right)^{e_i}$. For squarefree r these coincide; for $r = 63 = 3^2 \cdot 7$ the Jacobi symbol is $\left(\frac{\cdot}{7}\right)$ and for $r = 75 = 3 \cdot 5^2$ it is $\left(\frac{\cdot}{3}\right)$, and both of these rungs occur in our data. For prime r there is a single nontrivial real character and the distinction is vacuous; for composite r , testing only the Jacobi symbol is strictly weaker than testing all real characters. Note also that $\left(\frac{x}{r}\right) = +1$ does not mean x is a quadratic residue when r is composite, so the hypothesis must be phrased in terms of the character, not of residuosity.

The criteria simplify considerably once one uses $r \equiv 3 \pmod{4}$, and the simplification explains exactly which rungs can exhibit which levels.

Proposition 4.5 (criteria in reduced form). *Let rung r fail for p , and let $H \leq (\mathbb{Z}/r)^\times$ be as in Definition ?? . Recall $r \equiv 3 \pmod{4}$.*

1. *A real character $\chi \pmod{r}$ certifies the failure if and only if $\chi(-1) = -1$ (χ is odd) and $H \leq \ker \chi$. In particular the Jacobi symbol certifies if and only if $\left(\frac{\ell}{r}\right) = +1$ for every prime $\ell \mid q$. Once this holds, $\left(\frac{-q}{r}\right) = -1$ follows from $r \equiv 3 \pmod{4}$.*
2. *$-q \in H$ if and only if $-1 \in H$. Hence the support certificate of Definition ?? says exactly that $-1 \notin H$.*
3. *There are exactly $2^{\omega(r)-1}$ odd real characters mod r , one of which is the Jacobi symbol. Consequently a level-RC failure can occur only when $\omega(r) \geq 2$.*

Proof. Since q is a product of the primes generating H , we have $q \in H$; so if $H \leq \ker \chi$ then $\chi(q) = 1$ and $\chi(-q) = \chi(-1)$, giving (1). For the Jacobi symbol, $\left(\frac{-1}{r}\right) = (-1)^{(r-1)/2} = -1$ because $r \equiv 3 \pmod{4}$. Part (2) is the same observation: $-q \in H \iff -1 \in H$ because $q \in H$. For (3), $\chi_T(-1) = \prod_{i \in T} (-1)^{(p_i-1)/2}$, so χ_T is odd exactly when T meets $\{i : p_i \equiv 3 \pmod{4}\}$ in an odd number of elements; that set is nonempty since $r \equiv 3 \pmod{4}$, and the number of such T is $2^{\omega-1}$. When $\omega = 1$ there is therefore exactly one odd real character, which must be the Jacobi symbol, leaving no room for level RC. $\square$

Definition 4.6 (certificate hierarchy). Let rung r fail for p (so $\gcd(q, r) = 1$ by Lemma ??), and let $H \leq (\mathbb{Z}/r)^\times$ be the subgroup generated by $\{\ell \pmod{r} : \ell \mid q\}$. The failure has level

J if the Jacobi symbol $\left(\frac{\cdot}{r}\right)$ satisfies Corollary ??;

RC if some real character mod r does, but the Jacobi symbol does not;

S if no real character does, but $-q \pmod{r} \notin H$;

R (*residual miss*) otherwise: $-q \in H$, yet the divisors of q^2 —whose exponents are bounded—miss the class.

Proposition 4.7. *The levels are nested. Every J-certificate is an RC-certificate, and every RC-certificate implies $-q \notin H$. Level R is what remains after all local support information is exhausted.*

Proof. The first claim is Remark ?? . For the second, if χ is trivial on every $\ell \mid q$ then $H \leq \ker \chi$, while $\chi(-q) = -1$ puts $-q$ outside $\ker \chi$. $\square$

Levels RC and S are finer tests in principle. In particular, S detects obstructions from higher-order characters and from subgroups of index greater than 2 that no real character sees. Section ?? measures how often these finer tests are needed and where they first appear. Section ?? also shows that level S is confined to an explicit finite set of rungs.

For the smallest rung, level R is empty unconditionally, because modulo 3 there are only two characters.

Proposition 4.8 (rung 3 has no residual misses). *Let $p \equiv 1 \pmod{12}$ and $A = (p+3)/4$, so $r = 3$. Then rung 3 hits unless its failure is detected by a real character. More explicitly, it fails precisely when $3 \nmid A$ and every prime $\ell \mid pA$ satisfies $\ell \equiv 1 \pmod{3}$.*

Proof. If $3 \mid A$ take $u = 3$: then $3 \mid q$, $9 \mid q^2$, and both congruences of Lemma ?? read $0 \equiv 0 \pmod{3}$, so rung 3 hits. If $3 \nmid A$ then $\gcd(q, 3) = 1$ and (??) has two terms, $N_3(p) = \frac{1}{2}(\tau(q^2) + \chi(-q)\tau((q^2)_+))$ with χ the nontrivial character mod 3 (real, so conjugation is immaterial). If some $\ell \mid q$ has $\chi(\ell) = -1$ then $\tau((q^2)_+) < \tau(q^2)$ and $N_3 > 0$ either way. If every $\ell \mid q$ has $\chi(\ell) = 1$ then $\tau((q^2)_+) = \tau(q^2)$ and $\chi(-q) = \chi(-1) = -1$, so $N_3 = 0$; since $(\frac{\cdot}{3})$ is the only nontrivial real character mod 3, this is a level-J certificate. $\square$

4.1 Which rungs can carry a level-S failure

No real character detects a level-S failure. In the computation, every such failure occurs at the single rung $r = 51$ (§??). Higher-order characters still detect the support obstruction in the centered Fourier formulation. The concentration at $r = 51$ is explained by a constraint on H that every real character misses.

Lemma 4.9 (4 is already in the finite box). *For every rung under consideration, $4 \in B_{q,r}$ and hence $4 \in H$.*

Proof. $4A = p + r$ gives $pA^{-1} \equiv 4 \pmod{r}$. Because $\gcd(q, r) = 1$, the prime p does not divide A , so p and A are coprime divisors of $q = pA$. Their ratio p/A is one of the points of $B_{q,r}$ in Proposition ??. Thus $4 \in B_{q,r} \subseteq H$. $\square$

The same identity has a character-valued form which is useful for predicting how often the first certificate level occurs.

Corollary 4.10 (exact coset identity). *For every real character χ modulo r ,*

$$\chi(A) = \chi(p).$$

Proof. The congruence $4A \equiv p \pmod{r}$ gives $\chi(A) = \chi(p)\chi(4)^{-1}$, while $\chi(4) = \chi(2)^2 = 1$. $\square$

Any real χ has $\chi(4) = \chi(2)^2 = 1$, so every such character has $4 \in \ker \chi$. The condition cannot be detected by the character test in Corollary ??. It does, however, impose a strong restriction on the support subgroup.

Proposition 4.11 (when level S is possible). *Write $G = (\mathbb{Z}/r)^\times$. Rung r can carry a level-S failure only if some subgroup $H \leq G$ satisfies*

$$4 \in H, \quad -1 \notin H, \quad (-1)H \in (G/H)^2, \quad (5)$$

and any H satisfying (??) does produce one whenever it is realized as a support subgroup. In particular level S is impossible at every prime rung.

Proof. The first condition is Lemma ??, the second is Proposition ??(2), and the second also forces the rung to fail, since all divisors of q^2 lie in H . For the third: the real characters trivial on H are exactly the characters of G/H of order dividing 2, and by Proposition ??(1) the failure escapes every real-character certificate precisely when all of them take the value +1 at -1 . The intersection of the kernels of all order- ≤ 2 characters of a finite abelian group is its subgroup of squares, which is the stated condition.

If r is prime then G is cyclic of order $r - 1 \equiv 2 \pmod{4}$. Then $-1 \notin H$ forces $|H|$ odd, so G/H is cyclic of order $2m$ with m odd and $(-1)H$ is its unique element of order 2 — not a square, since $4 \nmid |G/H|$. No H qualifies. $\square$

Corollary 4.12. *Among $r \equiv 3 \pmod{4}$ with $r < 300$, level S is possible only for*

$$r \in \{51, 119, 123, 187, 195, 219, 255, 267, 287, 291\},$$

and 51 is the smallest by a wide margin. At $r = 51, 119$ and 123 the admissible subgroup is unique and cyclic of index 4; at $r = 51$ it is $H = \langle 2 \rangle = \{1, 2, 4, 8, 13, 16, 26, 32\}$ inside $(\mathbb{Z}/51)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/16$.

Dropping Lemma ?? from (??) would admit fourteen rungs below 140 alone,

$$15, 35, 39, 51, 55, 75, 87, 91, 95, 111, 115, 119, 123, 135,$$

with nothing to distinguish 51, and with $r = 15$ visited far more often than $r = 51$. The condition $4 \in H$ eliminates eleven of the fourteen. This also shows that the concentration at $r = 51$ is not simply an artifact of which rungs are reached: level-RC failures occur at $r = 51, 75, 91, 99, 115$. Thus these rungs are present in the data and can fail for local reasons. Nevertheless, at 75, 91 and 115 every subgroup that would otherwise satisfy (??) omits 4, and at 99 none exists in the first place.

This gives a simple check on any future computation: **a level-S failure can never occur at $r = 15, 35, 39, 55, 75, 87, 91, 95, 111, 115$ or 135.** The next rung at which one is possible is $r = 119$.

5 Exhaustive classification below 10^{15}

For each prime in $\mathcal{H}$, the computation follows the ladder to its first hit and classifies every preceding failure according to Definition ?. The resulting data set contains 932,642,160,749 primes and 557,964,077,974 failed rungs below 10^{15} . We computed the 10^{13} stage on CPUs and the extension to 10^{15} on a GPU. The GPU statistics and certificates are byte-for-byte identical to the CPU reference on every overlapping range. Independent exact-arithmetic and machine-word implementations agree on all overlapping ranges, and every fine certificate or deep-prime claim—all 585,677 of them—was checked again in exact arithmetic on a machine different from the one that produced it. Section ?? gives the algorithm and verification protocol.

Table 1: Observed maximum search depth $D(p)$, the first hitting rung, on $\mathcal{H}$. Long plateaus are followed by small record increments; they are evidence of an extreme tail, not of a fixed ceiling.

cutoff X	$\max_{p < X} D(p)$	first record holder in the census
10^6	59	806,521
10^7	107	8,803,369
10^8	107	8,803,369
10^9	107	8,803,369
10^{10}	107	8,803,369
10^{11}	107	8,803,369
10^{12}	111	654,730,707,409
10^{13}	131	6,367,024,012,441
10^{14}	131	6,367,024,012,441
10^{15}	155	172,538,390,619,841

The depth record remained 107 across three decades of new primes, then moved to 111, to 131, and—within the fifteenth decade—to 155, at

$$p = 172,538,390,619,841, \quad \frac{4}{p} = \frac{1}{43134597654999} + \frac{1}{48015316516502146586034030} + \frac{1}{567275091541152628215802033426900470},$$

and the displayed decomposition is exact. In the coprime-pair parametrization of Proposition ??, the corresponding pair is $d_1 = p$ and $d_2 = 288,157,589$; their sum is divisible by 155. The depth records only this first hit: among the 62 rungs in $[155, 400)$, 17 hit. Below 10^{15} the depths 111, 115, 119, 123, 127, 131 (eight primes), 135 (three), and 139 (one) all occur, while 143, 147, and 151 do not. Thus the record follows a gap, just as the earlier record at 107 did. Such long plateaus are consistent with an exceptionally thin tail. The numerical evidence suggests not a constant bound $D(p) \leq R$, but an unbounded envelope that grows too slowly to estimate reliably from record values alone. This remains a numerical interpretation, not a theorem; the current record is no more a bound than its predecessor was.

5.1 A numerical check of logarithmic record growth

Elsholtz asked whether the record might be modeled by $C \log X$ [?]. Table ?? gives ten cumulative observations, $X = 10^6, \dots, 10^{15}$. If the proposed N denotes the number of tested primes rather than the interval endpoint, then $\log N = \log X - \log \log X + O(1)$ by the prime number theorem; these ten record values cannot distinguish the two scales. We use the endpoint X , which is the quantity recorded by the census. A least-squares fit constrained to pass through the origin gives

$$M(X) \approx 4.531 \log X = 10.433 \log_{10} X, \quad (6)$$

with a root-mean-square error of 15.53 rungs. Allowing an intercept gives

$$M(X) \approx 38.13 + 7.055 \log_{10} X,$$

with root-mean-square error 11.83 and $R^2 = 0.746$. The constrained fit happens to predict 156.5 at 10^{15} , close to the observed 155, but this endpoint agreement is retrospective and should not be confused with a successful prediction.

As a simple forward check, fitting only through 10^{13} gives $M(X) = 4.664 \log X$ in the constrained model. It predicts 150.3 and 161.1 at 10^{14} and 10^{15} , compared with 131 and 155. The affine fit on the same training range predicts 132.6 and 138.8: it follows the next plateau but misses the jump to 155. Thus a logarithm gives a reasonable scale for the record, but one constant does not describe the plateaus and jumps visible in the data.

There is also a statistical limitation. These are cumulative maxima, so the ten points are strongly dependent; a record is repeated until a rarer prime appears. A genuine extreme-value model must start from the probability that a prime survives each rung and must retain the dependence between successive values $(p + r)/4$. The fixed-rung calculation below supplies one part of such a hazard model, but not yet the cross-rung joint law. We therefore regard $C \log X$ as a plausible descriptive scale, not an estimated growth law.

All failures at $r = 3$ are level J, as Proposition ?? predicts. Residual misses first appear at $r = 11$ and are concentrated at small rungs, while the finer local levels occur only on the thin subsequence of composite rungs. Their full counts and localization are reported below on the same population as the main census.

5.2 What the statistics say

The main empirical facts can be stated briefly. From 10^6 to 10^{15} , the cumulative residual share falls monotonically from 3.2634% to 0.8227%, although the residual count grows to 4,590,145,833. Residual misses are concentrated at divisor-poor shifted values, but a random-occupancy model for the reachable residue classes fails by orders of magnitude. The divisors of q^2 are too dependent for their count alone to predict a hit.

At a fixed squarefree rung, the unrestricted factor-support calculation is more successful. Selberg–Delange theory gives $N_r(Y) \sim c_r Y / \sqrt{\log Y}$ with an explicit Euler-product constant [?]. The exact identity $\chi_r((p+r)/4) = \chi_r(p)$ and the primality of $p = 4A - r$ supply corrections absent from an independent-integer model. After matching both effects, hard-class primes in a sample below 1.2×10^7 still have level-J rates 1.33–3.82 times those of the affine-prime control. This is a finite-range effect, not yet an asymptotic constant.

These calculations explain why the logarithmic fit in §?? is only a scale model. It would require a joint survival law across successive rungs, whereas the available analysis gives marginal hazards at six early rungs. The detailed constants, controls, and auxiliary tables are kept in the companion note [?] rather than repeated here.

6 Verification to $1.8446744065 \times 10^{19}$

The census asks how each hard-class prime fails before it succeeds. The complementary question — is every prime in a range solvable at all? — is answered far faster by running the arc of §?? in reverse: instead of following each prime down its ladder, apply the classical formulas algorithmically until only a vanishing residue of primes needs individual attention. The strategy — sweep the covering family’s parameters rather than search, extend the wheel far beyond the six classes modulo 840, and finish the thin residue with a direct solver that factors $(p+r)/4$ — was outlined to us by Elsholtz [?]. The two computations are orthogonal by design; in the covering identity $p+k=4ade$ the parameter k is a rung of size roughly p/m , some nine orders of magnitude deeper than the census ever searches, and covers are never applied to the census population, whose statistics they would contaminate.

6.1 Two certificate families and a derived wheel

Two families of class-level certificates drive the sieve. The first (F1) is the covering family (??) itself: reparametrized by $N = acd$ and $\alpha = c^2d$ it is exactly $\{(N, \alpha) : \alpha \mid N^2\}$ with modulus $m = 4N - 1$, and this completeness has a structural blind spot — the modulus is always $\equiv 3 \pmod{4}$, so F1 can never constrain a residue modulo 8. The second family (F2) closes the gap: *uniform rung certificates*, pairs (r, u) with $\gcd(r, u) = 1$ for which both divisor conditions of Lemma ?? are forced by the class of p modulo $L = 4\text{lcm}(r, u)$ alone, so that a single exact-arithmetic check certifies the whole class. Together the two families *derive* the classical reduction: applied to the classes modulo 840 they reproduce exactly Mordell’s six square classes, so Mordell’s theorem is demoted from an input to a regression check.

At the production modulus $M = 2,042,040$ the derivation kills 366,332 of the 368,640 unit classes — each with a named certificate recorded in a published table — leaving 2,308 surviving lanes, about one touched position per 885 integers. Nothing is transcribed from the literature (Terzi’s published class table is reported to contain small errors [?]); the table is regenerated at

startup by every scanner, which refuses to run on any digest mismatch, and re-derived independently by a dependency-free exact-arithmetic verifier.

6.2 The pipeline and its discipline

A window of integers passes through five stages. First, the wheel keeps only the 2,308 difficult residue classes. Second, 3,745 additional filters certify arithmetic progressions which do not fit into the wheel modulus. Third, deterministic Miller–Rabin separates primes from composites. The surviving primes then enter the rung walk of Lemma ??, which emits one witness divisor per prime. A direct search through (??) is retained as a final fallback. If both per-prime searches fail, the number is written as a *survivor* and the run fails.

The bulk stages certify 99.995101% of the wheel positions. Miller–Rabin leaves 156.377 billion primes; every one is solved by the rung walk, the direct fallback is never needed, and the survivor count is zero. The 6.26 TB deterministic certificate stream is committed shard by shard by SHA-256. Selected streams were reproduced byte for byte, and more than nine million certificates were checked in exact arithmetic, including an exhaustive window at the upper edge. The full audit is published with the run metadata rather than repeated here.

6.3 The 64-bit boundary, and the record

The natural ceiling of the implementation is not a power of ten. The direct solver probes $n = p + 4a^2d \leq p + 2048$ and factors it by methods whose integer square root is exact below $(2^{32} - 1)^2$; both scanners therefore refuse to start above

$$P_{\max} = (2^{32} - 1)^2 - 2049 = 18,446,744,065,119,614,976,$$

which is 8,589,936,640 below 2^{64} . Every modular product in the fast paths is checked against the frozen reference on adversarial inputs around 2^{63} . The scanners refuse an interval crossing $P_{\max}$ rather than allowing an arithmetic limit to appear as a mathematical survivor. The verification used 460 shards in two runs, split at 10^{19} .

The conclusion is: *every prime* $255,255 \leq p < P_{\max}$ *admits an Erdős–Straus decomposition*. The lower range is already covered by the censuses. Thus the conjecture is verified to about $1.8446744065 \times 10^{19}$, extending the previous record of 10^{18} [?] by a factor of about 18.4. The honest statement stops there. It is not a verification of the final 8,589,936,640 integers below 2^{64} and hence not of the full 64-bit range. The published class tables, filter tables, shard summaries, stream digests, and exact-arithmetic verifier allow any part of the computation to be regenerated without repeating the entire run.

7 The computation

To classify all failed rungs, we must completely factor every shifted value $A = (p + r)/4$ encountered before the first hit. Factoring these values independently is unnecessarily expensive; their arithmetic-progression structure permits a bulk sieve.

7.1 Bulk sieve-factorization

Within one of the six classes modulo 840, write $p = p_0 + 840j$. At a fixed rung,

$$A = \frac{p + r}{4} = A_0 + 210j, \quad A_0 = \frac{p_0 + r}{4}. \quad (7)$$

For every prime $\ell \nmid 210$,

$$\ell \mid A \iff j \equiv -A_0 \overline{210} \pmod{\ell}.$$

For each ℓ , the sieve updates exactly the entries divisible by ℓ . Sieving through $\sqrt{A_{\max}}$ leaves a prime cofactor, eliminating primality tests and Pollard rho from the main computation. Once only a thin set of primes remains on the ladder, the scanner switches back to per-value factorization. Against the naive implementation this gives speedups of $7.3\times$, $12.8\times$, and $20.3\times$ at 10^{10} , 10^{11} , and 10^{12} , respectively, with identical output.

7.2 Verification protocol

The optimized scanners (CPU and GPU) are checked against an independent exact-arithmetic reference and against the earlier per-value implementation. Repeated runs, thread counts, window sizes, and independent shardings give identical counts. The final shards (192 at 10^{13} , 400 at 10^{15}) are checked for gaps and overlaps, and every fine-certificate and deep-prime record—20,001 at 10^{13} , 585,677 at 10^{15} —is re-derived in exact arithmetic on a machine different from the one that produced it. The verifier additionally enforces Proposition ?? mechanically. If a claimed level-S failure occurs at an inadmissible rung, or if its support subgroup omits 4, the verifier reports a contradiction to the proposition rather than treating the record as a simple format error. Each reported solution is verified by the integer identity $4ABC = p(AB + BC + CA)$.

7.3 Identity sweeps and the quadratic obstruction

The census and a record-height verification solve different computational problems (§??): verification needs each integer covered once, by any identity; classification needs every failed rung preceding the first hit. The identity-first approach uses the classical Type I family of §?? (Mordell [?]; used as a large-scale presieve by Salez [?]): every positive value

$$n = (4acd - 1)k - 4a^2d, \quad a, c, d, k \in \mathbb{Z}_{>0}, \quad (8)$$

admits an Erdős–Straus decomposition. A finite parameter sweep is therefore an efficient way to cover many integers. However, this family can never produce a square.

Proposition 7.1 (no square occurs in the Type I family). *For positive integers a, c, d, k , the integer in (??) is not a perfect square.*

Proof. Put $M = 4acd - 1$ and suppose that $n = Mk - 4a^2d = s^2$. Since $4acd \equiv 1 \pmod{M}$,

$$n(2cd)^2 \equiv -4a^2d \cdot 4c^2d^2 = -(4acd)^2d \equiv -d \pmod{M}.$$

Thus $-d$ would be a square modulo M . But $\gcd(d, M) = 1$ and $(\frac{-d}{M}) = -1$. Indeed, $M \equiv 3 \pmod{4}$ gives $(\frac{-1}{M}) = -1$. For every odd prime $\ell \mid d$, quadratic reciprocity and $M \equiv -1 \pmod{\ell}$ give

$$\left(\frac{\ell}{M}\right) = (-1)^{(\ell-1)/2} \left(\frac{M}{\ell}\right) = (-1)^{(\ell-1)/2} \left(\frac{-1}{\ell}\right) = 1;$$

if $2 \mid d$, then $M \equiv 7 \pmod{8}$ and $(\frac{2}{M}) = 1$. Multiplication over the prime factors of d yields $(\frac{d}{M}) = 1$, a contradiction. $\square$

For the main Type I family, this proposition is an elementary instance of the quadratic obstruction behind the theorems of Schinzel and Yamamoto [?, ?]: no finite collection of polynomial congruence identities covers an entire quadratic-residue class. Prime inputs are not squares,

so this does not obstruct any one prime; it explains why a fixed finite identity sweep cannot replace the rung-by-rung census on $\mathcal{H}$.

Elsholtz also noted 288, 336, and approximately 9545 as small composite exceptions in an unrestricted parameter sweep [?]. We do not need an exhaustive claim about that composite exception set: the production sieve is a prime verification and starts at 255,255, while the lower range is handled independently. The relevant part of the suggestion is the change of enumeration order—generate the progressions from bounded (a, c, d) and apply them in bulk, then use a direct solver only on what remains—which is exactly the pipeline of §??.

8 Relation to prior work

The divisor-congruence criterion of Lemma ?? is classical and appears in several equivalent guises. Rav’s coprime-divisor criterion [?], as presented for example by Chen and Koo [?], is equivalent to it by Proposition ??; the centered exponent box is the same criterion after the change of variables $u/q = d_2/d_1$. Mordell’s identities [?] and the modular equations catalogued by Salez [?] are, in the present language, individual rungs: a polynomial identity valid on a residue class of p prescribes A (and hence r) together with a divisor u whose residue is guaranteed to be the target class. Yamamoto [?] and Schinzel [?] established the quadratic obstruction that Corollary ?? localizes to a single rung. Elsholtz and Tao [?] give the Type I / Type II taxonomy used in Proposition ??, together with mean-value theorems for the number of solutions; their framework subsumes the parametrization here. Vaughan [?] bounded the exceptional set by $\ll N \exp(-c(\log N)^{2/3})$. Prior to this work the conjecture was verified numerically to 10^{18} [?], after Swett [?] and Salez [?]; §?? extends the record to $(2^{32} - 1)^2 - 2049$. A recent preprint [?] obtains density one within $n \equiv 1 \pmod{4}$ using divisor conditions rather than congruence identities, and is the closest antecedent to the perspective taken here.

We therefore make no novelty claim for the divisor-lattice or coprime-pair criterion. Their ingredients are known. What we add is the certificate hierarchy of Definition ??, its exhaustive measurement, and the released certificates.

9 What remains

The local part is finite. At a fixed rung, the character expansion identifies every subgroup obstruction, and the census measures what remains when those obstructions vanish. The unresolved part is cross-rung: prove that no prime can miss the target class in every finite exponent box

$$B_{q_r, r}, \quad q_r = p(p+r)/4, \quad r = 3, 7, 11, \dots$$

The difficulty is that the group and the factorization both change with r . Selberg–Delange theory describes fixed character restrictions for typical integers, but primality and the affine relation $4A = p + r$ couple the prime factors of successive shifted values. Standard large-sieve or independent-hazard arguments do not retain that dependence.

A useful next result would be either a product-set theorem forcing $-1 \in B_{q_r, r}$ once the subgroup obstructions disappear, or a cross-rung cancellation estimate showing that the Fourier errors cannot all cancel their positive main terms. Even an almost-all version would test whether the observed plateaus and jumps arise from a slow extreme-value law. A uniform statement for every prime would prove the conjecture; this is why the remaining step is not a modest extrapolation of the census.

The second concrete question concerns level S. The group calculation says exactly which rungs can support it; below 10^{15} only $r = 51$ is occupied, although 119 and 123 are admissible and reached. Predicting the first occurrence at a second rung requires the missing distribution of support subgroups along $(p + r)/4$.

10 Conclusion

The certificate hierarchy explains almost every failed rung in the hard class, usually through the Jacobi symbol alone. What remains is small in proportion but not in count: the residual share falls to 0.8227%, while 4.59 billion residual misses remain below 10^{15} . Those misses occur disproportionately at divisor-poor shifted values, yet their divisors are far too dependent modulo r for a random-occupancy argument. The level that no real-character certificate detects is also highly concentrated. Because $4A = p + r$ puts 4 in every finite exponent box, and hence in every support subgroup, level S can occur at only ten rungs below 300. Every observed instance occurs at the smallest of them.

The depth data carries a separate message. The maximum first-hit rung stayed at 107 for a long interval and then rose through 131 to 155, with intermediate depths populated. This is the pattern expected from a very thin extreme tail. It provides no evidence for a uniform $O(1)$ depth bound. The suggested model $M(X) = C \log X$ gives $C = 4.531$ and the correct order of magnitude, but misses the observed record sequence by 15.53 rungs root-mean-square and does not reproduce its long plateaus. Our numerical expectation is that the record depth is unbounded but grows extraordinarily slowly; the range $p < 10^{15}$ is not enough to identify that growth law.

The fixed-rung analysis makes the limitation more precise. In an unrestricted model, the factor-support component of the level-J hazard has the explicit scale $c_r/\sqrt{\log X}$. In the actual ladder, $\chi_r((p + r)/4) = \chi_r(p)$ exactly, and the primality of p further changes the local factorization probabilities. Even after these effects are included in an affine-prime control, hard-class primes have level-J rates larger by factors 1.33–3.82 at the six rungs tested. These are now measured departures from the independent model, not a general warning that dependence may exist. What remains unknown is their behavior at height and their dependence across rungs.

None of this proves the conjecture. A vanishing share of residual failures does not prevent one prime from missing every rung, and an observed record is neither an upper bound nor a theorem of divergence. The useful outcome is the separation itself: local obstructions are finite and certifiable, while the remaining problem concerns the structured divisors of $p^2((p + r)/4)^2$ along shifted prime values. That is the part a proof must control.

The verification of §?? puts a computational floor under that separation. Every prime below $(2^{32} - 1)^2 - 2049$ is solvable; the small initial range is covered by the censuses and the verification sieve covers the rest. Equivalently, the combined computation reaches $1.8446744065 \times 10^{19}$ with zero survivors. The class-level certificates are published directly, while the per-prime streams are hashed and can be regenerated shard by shard. The final 8,589,936,640 integers below 2^{64} remain outside the claim. The two products therefore meet in a narrower but useful statement: the verification shows solvability throughout the computed range, while the classification says which part of that solvability is not explained by local arithmetic.

Data and code availability

The complete repository is <https://github.com/Attitude-Adjuster/erdos-strauss>. Questions, corrections, code problems, and independent reproduction reports may be submitted at <https://github.com/Attitude-Adjuster/erdos-strauss>.

The repository contains the exact-arithmetic reference implementation, the independent C++ scanners (the frozen reference and an optimized variant proved output-identical against it), the CUDA scanner, the certificate verifier—including the mechanical check of Proposition ??—the per-shard summaries of both censuses (192 shards at 10^{13} , 400 at 10^{15}), the merged depth and level histograms, the 20,001 fine-certificate and deep-prime records of the 10^{13} census, the 585,677 of the 10^{15} census, and the verification transcripts. The analysis scripts reproduce the cumulative and disjoint-decade statistics. Level-J failures are summarized rather than stored individually because each is regenerated directly from (p, A, r) .

For the verification sieve of §??, the repository additionally contains the certified class tables at both wheel moduli (surviving classes plus a named killing certificate for every other unit class, with digests), the pruned filter tables, both sieve implementations (the frozen reference and the optimized scanner, byte-identical on shared ranges), the independent exact-arithmetic verifier for the covering identity and class tables, and all 460 metadata records from the two record-height runs. These records contain the interval, counters, provenance, and SHA-256 commitment for each certificate stream. The repository also contains the re-derived shard edges and samples, the complete certificate set for the smaller verification window, and summaries and digests for the exhaustive window at the upper edge of the computation.

Each solution certificate is a triple (p, A, u) and is checked through $4ABC = p(AB + BC + CA)$. The per-shard summaries are published so that small pieces of the census can be reproduced independently rather than requiring the full run.

Direct enumeration also checks the conjugated character formula on the small examples used during development; the full table is retained in the source and repository tests.

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A detailed critical reading identified the composite- r gap in the original obstruction statement and sharpened the discussion of the analytic obstacles. It also supplied independent divisor counts used to check (?). Christian Elsholtz [?] independently supplied the coprime-divisor formulation of Proposition ?? together with its group-theoretic reading, verified the depth-155 example, cautioned that Terzi’s published class table contains small errors, and outlined the verification strategy that became §?? and §??: sweeping the covering family’s parameters, extending the wheel beyond the classes modulo 840, and resolving the remaining exceptions with a direct solver. Those comments prompted the explicit correspondence between the classical criterion, the exponent box, and the support subgroup, as well as the clarification that depth records only the first hitting rung.

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